

STRIVE

Detection Handoff

What the platform provides, and how to plug a real detector into it

Problem Statement	SIH26104
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Team	STRIVE, IISER Bhopal
Audience	Detection / ML teammates

One-line summary. The streaming platform, channel intelligence, codec robustness lab and evaluation harness are built and measured. The detector is a DSP placeholder scoring **AUC 0.708** on real ASVspoof data. Everything you need to replace it is described in §2; the corpus and GPU are already provisioned.

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1 Where the project stands

1.1 Built and measured

Streaming and real-time

Browser microphone → canonical 16 kHz mono → 20 ms binary PCM (pcm-v2) → independent server ingestion and scoring. Configurable window/hop geometry (2 s/1 s, 2 s/0.5 s, 4 s/0.5 s all tested). Multi-rate branch scheduler. Bounded capture queue so inference stalls cannot block audio. Optional Rust audio runtime. Per-stage latency instrumentation throughout.

Channel intelligence

Per-window occupied bandwidth, SNR proxy, clipping ratio, RMS dBFS, cross-window discontinuity and a conservative reliability scalar. Verified to separate G.711, Opus and narrowband transport from clean audio.

Codec robustness lab

Fifteen registered conditions (Opus 6/12/16/24/32 kbps, G.711 μ -law and A-law, 8 kHz narrowband, noise at 20/10/5 dB, packet loss 1/3/5%). Paired manifests that keep `source_id` across every derived clip so no split leakage is possible.

Evaluation harness

Streaming WAV simulator (deterministic and wall-clock modes), latency benchmark, and a codec benchmark producing EER, ROC-AUC, TPR at fixed FPR, Brier, score distributions and per-condition latency. **183 tests pass, 1 skipped.**

Not built: the detector

The default extractor is `dsp-surrogate-v1`: hand-designed spectral features, no learned weights, explicitly flagged `demo_only: true` on every event. Research adapters for the frozen NII countermeasure exist but **have never been run** — no model bundle is installed. Every detection number in this document comes from the placeholder.

1.2 Measured detection floor

ASVspoof 2019 LA, **unseen-generator** evaluation: reference index built on attacks A01–A06, evaluated on A07–A19. 20 vs 67 speakers, disjoint. 3,000 evaluation clips per condition, $n_0 = 1451$ genuine, zero failures across 18,000 evaluations.

Condition	EER %	Δ EER	AUC	Δ AUC	TPR@1%FPR
clean	35.04	—	0.708	—	0.056
opus_32k	36.84	+1.80	0.672	−0.036	0.070
opus_16k	37.44	+2.40	0.676	−0.031	0.046
g711_ulaw	37.86	+2.82	0.669	−0.039	0.075
narrowband_8k	38.07	+3.03	0.670	−0.038	0.072
noise_10db	48.43	+13.39	0.533	−0.174	0.103

Read this as a floor, not an achievement. AUC 0.708 is weak but non-trivial signal from unsupervised DSP features. It is the number a trained countermeasure must beat, and it is published precisely so that any improvement is measurable rather than asserted.

1.3 Measured latency

Live loopback WebSocket, 400 packets, 2s window / 0.5s hop, i5-1235U CPU.

Measurement	Normal	1200 ms injected inference
Packets acknowledged	400/400	400/400
Ingestion p95	0.615 ms	0.612 ms
Window-end → client p95	82.53 ms	1720.98 ms
Windows scored / discarded	13 / 0	6 / 7
Maximum pending depth	1	1

What the overload column buys you

With inference deliberately stalled at 1.2s per window, ingestion stays at 0.612ms and every packet is still acknowledged. Stale windows are discarded, not queued, and pending depth never exceeds one.

Practical consequence for you: a slow model degrades the update rate, it does not break the call. You are free to try a heavy model without first optimising it. Compute, queue, audio-lag and dropped-window counters make the cost visible in every event.

2 The detector interface

This is the whole contract. Anything satisfying it drops into the running system with no other changes.

2.1 What you must provide

```
class YourExtractor:
    id: str                # e.g. "nii-mms300m-⟨hash12⟩"; appears in every event
    is_surrogate: bool     # False for a real model -> removes the demo_only flag

    def extract(self, waveform: np.ndarray) -> Features:
        """Exactly window_s * 16000 float32 samples, normalised to [-1, 1]."""

    # Optional
    pipeline_version: str   # provenance beyond the model hash
    def identify_language(self, audio) -> tuple[str, str]: # (tag, source)
    def close(self) -> None # release GPU memory
```

2.2 What you must return

```
@dataclass
class Features:
    cm: np.ndarray          # countermeasure embedding, unit-normalised
    profile: np.ndarray     # speaker/acoustic profile vector, own dimension
    segments: list[Segment] # time-aligned sub-window vectors
    cm_score: float | None  # direct spoof probability in [0,1], or None
    metadata: dict          # free-form; surfaced in the event's features field

@dataclass
class Segment:
    start_s: float          # relative to window start
    end_s: float            #
    vector: np.ndarray      # all segments in a window must share a dimension
    token: int = -1         # phoneme/cluster id, -1 if unavailable
```

Field	Consumed by
cm	Track 1, FAISS retrieval against the labelled reference index. Must match the index's dimension and the extractor id it was built with.
profile	Track 2, in-call session consistency. Dimension may differ from cm and must stay fixed within a call.
segments	Track 3, boundary coherence across the window seam.
cm_score	Recorded as cm_head_score; not yet fused. See §4.

Three contracts that will fail loudly if broken

1. **Input shape is exact.** `extract` receives precisely `window_s × 16000` samples. Not approximately.
2. **Dimensions are fixed within a call.** Changing `profile` or `segment` dimension mid-call raises; the session store cannot mix vector spaces.
3. **The index must match the extractor.** Startup refuses if `index.metadata["extractor_id"] != extractor.id`, and research mode refuses a `demo_only` fixture index. This is deliberate: a silently mismatched index produces confident nonsense.

2.3 Where it is called

```
# strive/engine.py
features = self.extractor.extract(window.samples)          # per window
self.language, src = self.extractor.identify_language(...) # once, after
bootstrap
```

Failures are caught and reported as `MODEL_OR_INDEX_ERROR` with explicit uncertainty — never as a zero score. Set `raise_model_errors=True` in config while debugging.

3 Findings that should shape your design

These come from the surrogate, but the mechanisms are model-independent.

3.1 Noise destroys detection; codecs merely dent it

Every codec leg costs 0.031–0.039 AUC. Additive noise at 10 dB SNR costs **0.174** — about five times worse — driving AUC to 0.533 and collapsing the bonafide/spoof median gap to **exactly zero**.

Implication: noise robustness deserves more of your effort than codec robustness. Test under additive noise early; a model tuned only on clean or codec-degraded audio will look far better than it is.

3.2 The evidence gate is inverted under noise

Clips clearing the live-call evidence gate: 114 clean, **651** at 10 dB SNR — a 5.7× increase. The cause is the demo-mode energy VAD (`strive/audio.py::activity`, `RMS > 0.008`), whose voiced fraction rises from 0.564 to 1.000 under noise. Every frame is counted as speech.

The system gets more confident exactly when it should get less

This is safety-relevant and currently unfixed. Research mode substitutes a trained WebRTC VAD, which should behave better — **but that has not been measured**. Verifying it is a concrete, valuable early task.

3.3 The surrogate’s signal is generator-agnostic

Seen-generator AUC was 0.707; unseen-generator is 0.708. No drop. The placeholder is not memorising attack families — it is picking up a generic synthesis cue.

Implication: a trained countermeasure will probably show a real seen/unseen gap, because it *can* overfit to known attacks. Always report unseen-generator numbers. The corpus splits make this automatic: A01–A06 for the index, A07–A19 for evaluation.

4 How to move forward

4.1 Already provisioned

Resource	Status
GPU	stork (172.28.79.149): RTX 3090, 24 GB, 61 GB RAM, 32 threads
Corpus	ASVspoof 2019 LA complete, 122,299 files, md5-verified, at /media/storage/moonlab/strive-data/asv2019
Manifests	Official speaker-disjoint splits already built: reference 25,380 / validation 24,844 / test 71,237
Codec matrix	18,000 degraded clips across 6 conditions, ready to reuse
Python	3.12 venv for the app; system 3.10 for the Fairseq export

The Python 3.10 on *stork* is not an inconvenience — the NII export needs legacy Fairseq, which will not install alongside the 3.12 application runtime.

4.2 Suggested order

Step 1 — Get one real number. Read docs/MODELS.md, then:

```
python scripts/prepare_models.py --download --variant mms-300m --include-language
# separate Python 3.10 environment for the Fairseq export
python scripts/export_nii.py --variant mms-300m \
    --weights models/nii-source/model.safetensors --output models/cm.pt
# seal the bundle: manifest.json pinning sha256 for every file
python scripts/build_index.py data/manifests/asv2019/reference.csv --models models
python scripts/start.py --research
```

Start with **MMS-300M**, not XLS-R-2B. It is 1024-dimensional and far lighter; get the pipeline working before scaling the model.

Step 2 — Re-run the benchmark and compare against the floor.

```
python scripts/codec_benchmark.py \
    --matrix data/manifests/asv2019/matrix_full.csv \
    --reference-manifest data/manifests/asv2019/ref_sample.csv \
    --output evidence/codec-benchmark-research.json
```

Identical cohort, identical conditions. The delta against AUC 0.708 is the contribution, and it is directly quotable.

Step 3 — Fuse cm_score. The research extractor returns a direct spoof probability that is currently *recorded but unused*; Track 1 relies on retrieval alone. Wiring it in is likely the single largest accuracy gain available and needs no new model.

Step 4 — Channel-aware fusion (FUS-03). Under `noise_10db` the channel layer reports `snr_db = 19.6` and `quality = 0.898` while AUC collapses to 0.533. The signal needed to discount that evidence is already computed and already in every event — the fusion simply ignores it. Constants belong in config; the feature needs an on/off flag.

Step 5 — Prosody (BR-04) and identity (BR-05) branches. The `BranchResult` schema and multi-rate scheduler already exist. Both are optional by construction: fusion masks unavailable branches and renormalises.

4.3 Rules we have been holding to

1. **Never tune a threshold to make a demo look better.** Two known weaknesses (EMA warm-up, mature-call dilution) are pinned by tests on purpose.
2. **Absent evidence is masked, never zeroed.** A missing or stale branch must not contribute a score that reads as “genuine”.
3. **Unavailable measurements are null, never 0.**
4. **No fabricated numbers.** If a cohort is too small, the tooling says so — `low_fpr_sample_warning` exists for exactly this reason.
5. **Report unseen-generator results.** Seen-generator numbers flatter the model and do not predict field behaviour.

5 Open tickets

Ticket	Priority	Description
—	P0	Run research mode end to end; produce one real detection number
FUS-03	P0	Channel-aware reliability weighting (evidence is in hand)
—	P0	Fuse <code>cm_score</code> into Track 1
DEMO-01/02	P0	Single-command startup, deterministic demo scenarios
BR-04	P1	Minimal prosody branch
CODEC-05	P1	Automated $\Delta\text{EER}/\Delta\text{AUC}$ table (deltas are hand-computed)
CODEC-06	P1	Embedding preservation study; needs frozen encoders
UI-03/04	P1	Branch and channel explanations, live latency indicator
BR-05	P2	Identity branch adapter

Housekeeping before submission

- README badges are stale: they read `tests-170_passed` (now 183) and `p95_latency-6.7ms`. The 6.7 ms figure is *engine-only*; live transport p95 is 82.53 ms. A reviewer who notices will discount other claims.
- `native/audio-runtime/target/` (Rust build artefacts) and `data/manifests/` (absolute paths into stork) must be gitignored.
- Two commits are unpushed and roughly 28 files uncommitted.

6 Honest summary for judging

Strong. Streaming architecture with measured overload behaviour; channel intelligence; codec robustness on a real corpus with unseen attacks; privacy by construction; no mandatory paid APIs; every claim backed by a checked-in artefact.

Weak. The detector. Every metric is a placeholder at $AUC \approx 0.71$ clean and 0.53 under noise. The repository states this everywhere rather than obscuring it.

The single highest-leverage move

Get research mode running on stork's RTX 3090. Corpus, manifests, codec matrix, GPU and evaluation harness are all in place. What is missing is one model bundle and one benchmark run — and that converts “a measurable streaming system” into “a measurable streaming system *that detects voice clones*”.

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